USER GUIDE · ISTEMAAL AUR TESTING · AASAN ROMAN URDU

SENTINEL

Fire & Emergency Monitoring System

Aam istemaal karne wale ke liye qadam ba qadam: install, roz chalana, login, dashboard ka har hissa, aur har feature ka test, ke screen par kya dikhna chahiye. Saath mein masle aur un ka hal, hifazati usool aur print karne wali checklist.

Install
Ek dafa: software, .env, database, AI models

Chalana
3 windows: API · Dashboard · AI service

9 Tests
Aag demo · AI photo · sensors · sprinklers · naqsha · video

Hifazat
Safe Test Mode: test mein asli SMS/Email band



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1. Ye Guide Kis Ke Liye Hai?

Ye guide **aam istemaal karne wale** ke liye hai, programmer hona zaroori nahi. Is mein qadam ba qadam likha hai:

1. Sentinel ko pehli dafa computer par kaise **install** karein.
2. Roz kaise **chalayein** (start) aur band karein.
3. Website (dashboard) ka **har hissa** kya karta hai.
4. Har feature ko **khud test** kaise karein, aur test ke baad screen par **kya dikhna chahiye**.
5. Masla aaye to **kya karein**.

Sentinel ek line mein

Sentinel building ka **"smart chowkidar"** hai. Camera mein AI aag, dhuwan aur insaan pehchaanta hai. Sensors garmi aur dhuwan napte hain. Khatra ho to screen par alarm aata hai, SMS/Email/Push jaate hain, sprinklers chal sakte hain, aur naqshe par **mehfooz raasta** dikhta hai.

Guide kaise parhein

- `Is tarah ke dabbe` mein jo likha hai, wo **computer mein type** karna hai (ya copy-paste).
- ✅ = test pass hone par yehi dikhna chahiye.
- ⚠️ = dhyan dein, zaroori baat.
- `<project>` ka matlab hai project ka folder, is computer par: `C:\Users\m_fahad\Desktop\Project`.

3 hifazati usool (pehle parh lein)

> ⚠️ **1.** Ye system **test aur demo** ke liye hai. Ise asli fire alarm ki jagah **na** lagayein. Asli building mein smoke detectors aur fire panel zaroori hain.
>
> ⚠️ **2.** Test se pehle **"Safe Test Mode"** (Chapter 4) zaroor karein, warna test ke dauran **asli SMS aur Email** chale jayenge.
>
> ⚠️ **3.** Kisi **asli sprinkler, alarm ya camera** ko bina ijazat aur bina safe tareeqe ke system se na jorein.

2. Pehli Dafa: Zaroori Software Install Karna

Ye kaam **sirf ek dafa** karna hai. Har software ki official website se download karein.

#	Software	Kahan se	Kyun
1	**.NET 8 SDK**	dotnet.microsoft.com/download	Backend (API) chalane ke liye
2	**Node.js** (20.19 ya naya)	nodejs.org	Website (dashboard) ke liye
3	**Angular CLI**	Node ke baad command se (neeche)	Website chalane ka tool
4	**SQL Server** (Express ya Developer, free)	microsoft.com/sql-server	Database
5	**dotnet-ef**	Command se (neeche)	Database tables banane ka tool
6	**Python 3.13**	python.org (install mein "py launcher" tick rehne dein)	AI service

Angular CLI aur dotnet-ef install (PowerShell kholein):

```
npm install -g @angular/cli
dotnet tool install --global dotnet-ef
```

Check karein ke sab install ho gaya (har line alag chalayein):

```
dotnet --version        # 8.x ya naya
node --version          # v20.19 ya naya
ng version              # Angular CLI ka version
dotnet ef --version     # koi bhi version
py -3.13 --version      # Python 3.13.x
```

✅ Har command ek version number dikhaye, error nahi.

PowerShell kaise kholein: Start menu → "PowerShell" likhein → **Windows PowerShell** par click.

3. Pehli Dafa: Project Ka Setup

Ye bhi **sirf ek dafa**.

Qadam 1: `.env` file banana (settings aur passwords)

1. Project folder kholein: `C:\Users\m_fahad\Desktop\Project`.
2. `.env.example` file ki copy banayein aur naam `.env` rakhein (is computer par `.env` pehle se bani hui hai).
3. `.env` ko Notepad mein kholein aur ye bharein:

Line	Kya likhna hai
`Jwt__SigningKey=`	Lambi random chaabi (neeche command se banayein)
`Device__ApiKey=`	Ek aur random chaabi (machines ka password)
`Seed__AdminEmail=`	Login email, jaise `admin@sentinel.local`
`Seed__AdminPassword=`	Apna login password. Khali chhoda to API pehli dafa ek random password **terminal mein print** karegi

Random chaabi banane ki command (PowerShell):

```
[Convert]::ToBase64String((1..48 | % {Get-Random -Max 256}))
```

Jo text aaye, copy karke `=` ke baad paste karein. `Device__ApiKey` ke liye dobara chalayein.

> ⚠️ `.env` mein passwords hain. Ise **kisi ko na bhejein**, na internet par daalein.

Qadam 2: Database banana

```
cd C:\Users\m_fahad\Desktop\Project\backend
dotnet restore
dotnet ef database update --project Sentinel.Api
```

✅ Aakhir mein **"Done."** likha aaye. Is se SQL Server mein `SentinelDb` database aur saari tables ban jaati hain.

Qadam 3: Website ke packages

```
cd C:\Users\m_fahad\Desktop\Project\frontend
npm install
```

✅ Kuch minute baad bina laal "ERR!" ke khatam ho.

Qadam 4: AI service ka setup

```
cd C:\Users\m_fahad\Desktop\Project\ai-service
py -3.13 -m venv .venv
.\.venv\Scripts\python -m pip install -r requirements.txt
.\.venv\Scripts\python tools\download_models.py
```

- `download_models.py` **person (insaan) model** official website se laata hai aur us ka fingerprint (sha256) check karta hai.
- **Fire/smoke model** (`fire_smoke_yolox.onnx`) is computer par `ai-service\models\` mein pehle se hai. Naye computer par is file ko yahan copy karna hoga.

✅ `ai-service\models\` folder mein ye 3 files hon: `models.json` , `yolox_tiny.onnx` , `fire_smoke_yolox.onnx` .

4. Safe Test Mode (Test Se Pehle Zaroor)

Masla: is computer ki `.env` mein **asli** Twilio (SMS) aur Email ki settings hain. Test mein aag ya emergency bani to system **asli logon ko asli SMS/Email** bhej dega.

Hal: test ke dauran SMS aur Email band karna

1. `.env` Notepad mein kholein.
2. In lines ke shuru mein `#` laga dein (`#` wali line system parhta hi nahi):

```
#Notifications__Sms__TwilioAccountSid= ...
#Notifications__Sms__TwilioAuthToken= ...
#Notifications__Sms__FromNumber= ...
#Notifications__Sms__Recipients= ...
#Notifications__Email__SmtpHost= ...
```

3. Save karein. API chal rahi ho to band karke dobara chalayein (Chapter 5).

Is ke baad kya hoga:

- SMS: Live Event Log mein "**skipped (not configured)**". Koi SMS nahi jayega.
- Email: asal mein nahi jayega. Us ka poora text **API wali PowerShell window** mein print hoga ("console mode"). Test ke liye ye behtareen hai.
- Browser Push: sirf usi browser par jahan aap ne khud **Enable Alerts** dabaya ho.

Test khatam hone ke baad `#` hata dein taake asli alerts dobara chalein.

5. Roz Ka Kaam: System Chalana

Har hissa **alag PowerShell window** mein chalta hai. Windows band na karein jab tak system istemaal ho raha ho.

Window 1: Backend (API)

```
cd C:\Users\m_fahad\Desktop\Project\backend
dotnet run --project Sentinel.Api --launch-profile http
```

✅ Ye line aaye: `Now listening on: http://localhost:5291`

Window 2: Website (Dashboard)

```
cd C:\Users\m_fahad\Desktop\Project\frontend
ng serve
```

✅ Ye line aaye: `Local: http://localhost:4200/`

Window 3: AI Service

```
cd C:\Users\m_fahad\Desktop\Project\ai-service
.\.venv\Scripts\python -m sentinel_ai
```

✅ Ye line aaye: `models loaded: ['person', 'fire_smoke'], missing: []`

Window 4 (sirf demo ke waqt): Simulator

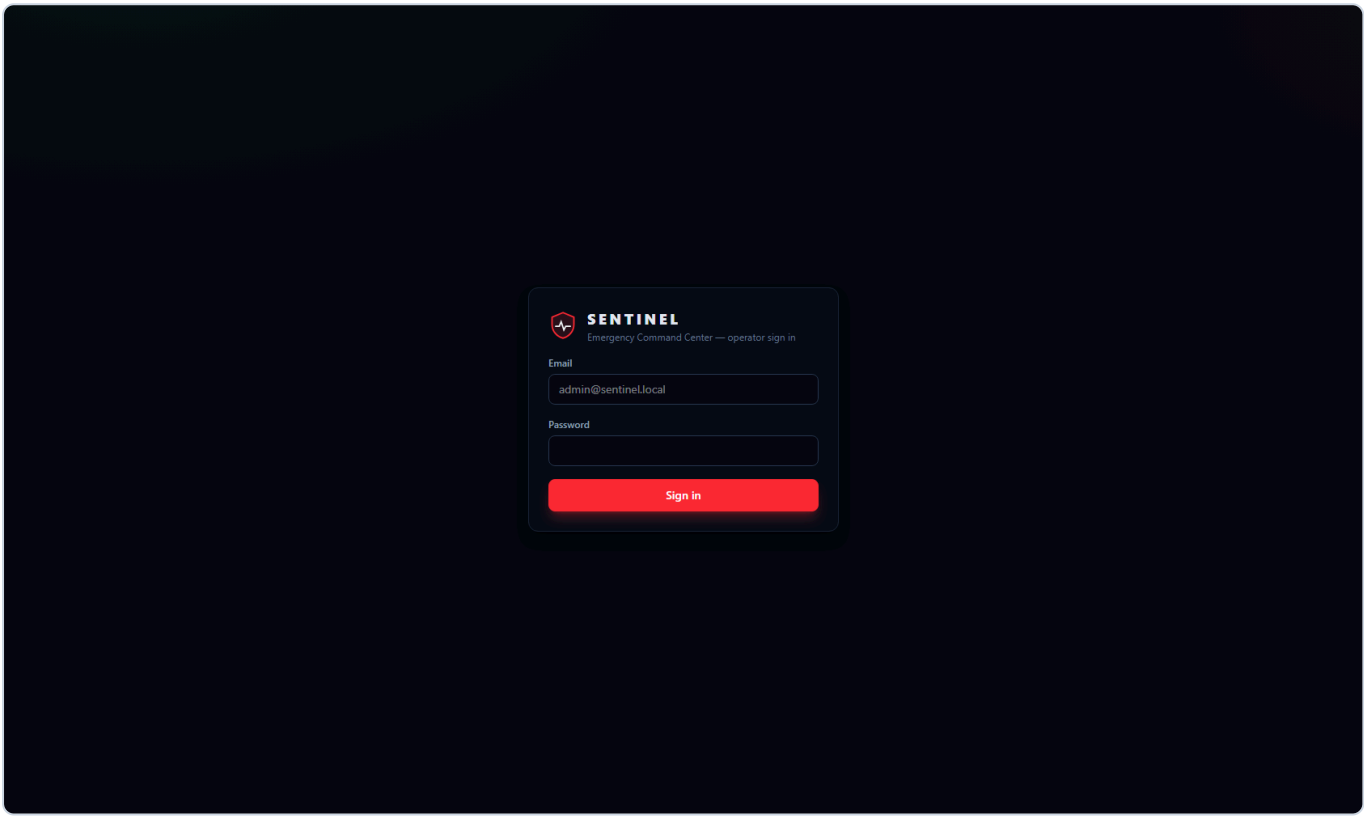
Simulator **nakli** cameras, sensors aur aag banata hai, taake asli aag ke baghair demo ho sake (Chapter 8).

System band karna

Har window mein **Ctrl + C** dabayein, ya window band kar dein. Data database mein mehfooz rehta hai.

6. Login Karna

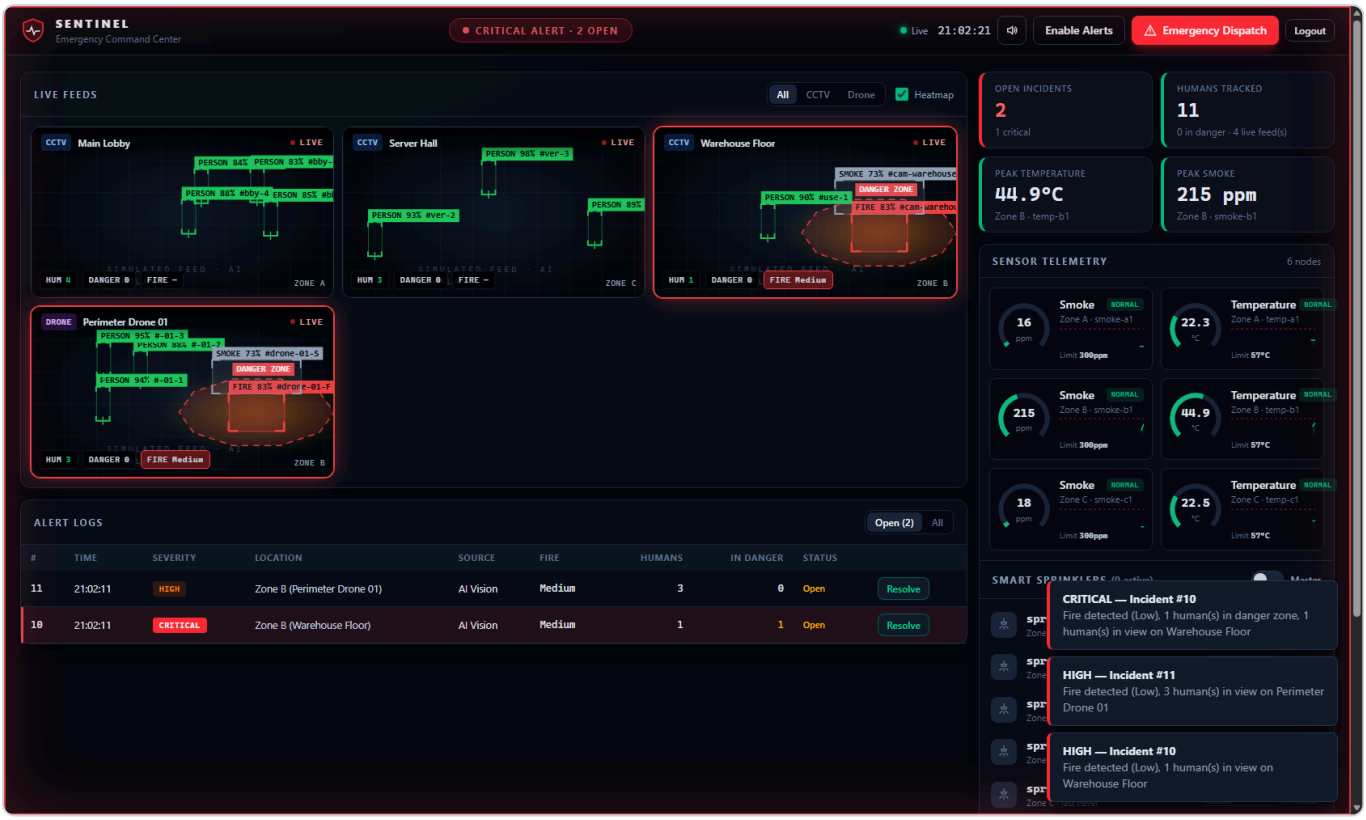
1. Browser (Chrome/Edge) kholein: **http://localhost:4200**
2. **Email:** `.env` wala `Seed__AdminEmail` (jaise `admin@sentinel.local`).
3. **Password:** `.env` wala `Seed__AdminPassword`. Khali chhoda tha to pehli dafa API ki window mein print hua tha.
4. **Sign in** dabayein.



✅ Dashboard khul jaye aur upar dayen taraf **Live** (hara nukta) likha ho.

Ghalat password: laal message aayega. Login 8 ghante tak chalta hai, phir dobara login.

7. Dashboard Ka Tour (Har Hissa Kya Hai)



Upar ki patti (Top bar)

Cheez	Matlab
Dashboard · Floor Plans · AI	Teen pages ke beech jana
Beech ka pill	Hara "ALL SYSTEMS NOMINAL" = sab theek. Laal "CRITICAL ALERT" = khatra
Live / Offline	Server se live connection hai ya nahi
Ghari	Abhi ka waqt
🔊 **button**	Alarm ki awaaz band/chalu
Enable Alerts	Is browser par Push notifications on karna
Emergency Dispatch (laal)	Haath se emergency elaan (Chapter 12)
Logout	Bahar nikalna

Dashboard ke hisse

Hissa	Kya dikhata hai	Aap kya kar sakte hain
KPI patti (4 dabbe)	Khule incidents, insaan, sab se zyada garmi, sab se zyada dhuwan	Sirf dekhna

Hissa	Kya dikhata hai	Aap kya kar sakte hain
Live Feeds (cameras)	Har camera, AI ke boxes aur zones	All/CCTV/Drone filter, Heatmap on/off, camera par click = bara
Sensor Telemetry	6 sensors ke gauges (dhuwan ppm, garmi °C)	Dekhna
Smart Sprinklers	Har sprinkler ON/OFF, Auto/Manual	Switch, Auto/Manual, Master
Alert Logs	Incidents ki list	Open/All filter, **Resolve**
Live Event Log	Har waqia (SMS gaya, sprinkler chala…)	Dekhna
Incident History & Logs	Aag ka graph aur history	Incident aur event type ka filter
Evacuation Route (estimate)	Naqsha aur mehfooz raasta	Naqshe par click = raasta

Camera tile par rang

Rang	Matlab
🔴 Laal dabba (dhadakta)	Aag
⚪ Gray dabba	Dhuwan
🟢 Hara dabba	Mehfooz insaan
🟠 Amber "IN DANGER"	Insaan khatre ke zone mein
🔴 Laal dashed shakal	Danger zone
🟢 Hari shakal	Safe zone (assembly point)
SIM DATA chip	Data nakli (test) hai
NO AI DATA	AI se 1.5 second se data nahi aaya

Severity (khatre ka darja)

Low (halka) → **Medium** → **High** (zyada) → **Critical** (sab se khatarnak).

8. Test 1: Aag Ka Demo (Simulator)

Maqsad: dekhna ke aag lagne par poora system kaise react karta hai.

> ⚠️ Pehle **Safe Test Mode** (Chapter 4) karein. Is demo mein insaan danger zone mein aate hain, jo **emergency** hai, aur SMS/Email chalte hain.

Qadam:

1. Window 1 (API) aur Window 2 (Website) chal rahi hon. Login karein.
2. `.env` se apni `Device__ApiKey` copy karein.
3. Nayi PowerShell window (Window 4):

```
cd C:\Users\m_fahad\Desktop\Project\backend
dotnet run --project Sentinel.Simulator -- --device-key <yahan Device__ApiKey> --ignite "Zone B"
```

4. Browser mein Dashboard dekhein.

✅ **Kya dikhna chahiye (1–2 minute mein):**

- [ ] Cameras par boxes hilte hue. **SIM DATA** chip.
- [ ] **Warehouse Floor** (Zone B) camera par laal aag ka dabba aur laal danger zone, hari "assembly point".
- [ ] Upar pill laal: **CRITICAL ALERT**. Alarm ki awaaz.
- [ ] **Alert Logs** mein naya incident, Location "Zone B", Severity High/Critical.
- [ ] **Live Event Log** mein lines: incident khula, SMS "skipped", email (API window mein print).
- [ ] Zone B ke garmi/dhuwan gauges barhte hue.
- [ ] Camera (AI Vision) wale incident par sprinklers **khud nahi** chalte: Live Event Log mein "sprinkler auto-activation skipped… SIMULATED".
- [ ] ⚠️ Lekin simulator nakli **sensors** ki readings bhi bhejta hai, aur un par SIMULATED ka nishaan **nahi** hota. Zone B ke sensors Alarm par pahunchein to Zone B ke **Auto sprinklers chal jaate hain** (sirf database mein, koi asli hardware nahi). Test ke baad unhein OFF karein (Chapter 11).

Band karna: Window 4 mein **Ctrl + C**.

Simulator ke options:

Option	Kaam	Default
`--ignite "Zone A"`	Us zone mein foran aag (kai dafa likh sakte hain)	—
`--fps 2`	Har camera ke frames per second	5
`--fire-probability 0`	Khud se random aag band	0.004
`--telemetry-interval 5`	Sensors kitne second baad	2

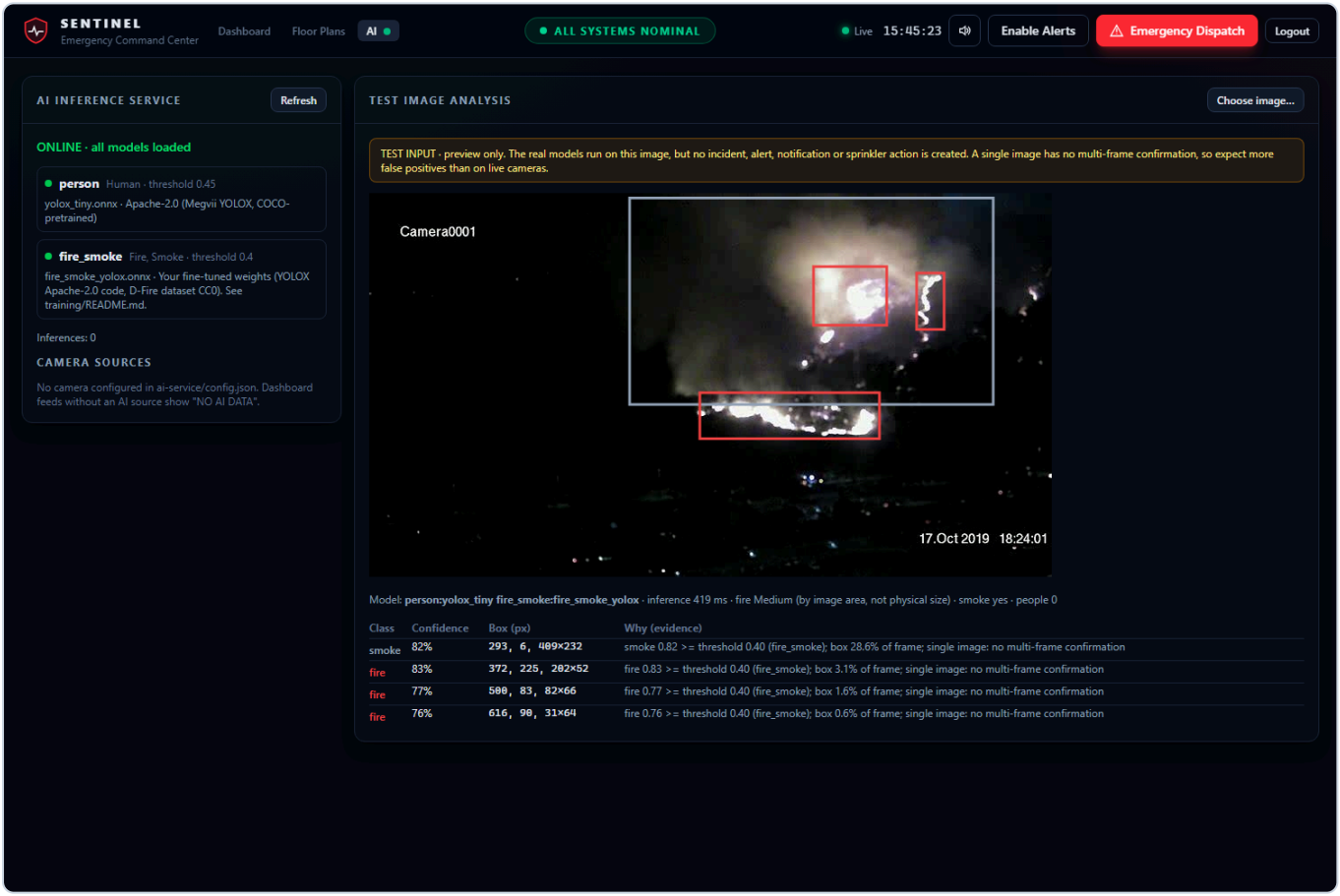
9. Test 2: AI Ko Photo Dikhana (AI Page)

Maqsad: dekhna ke AI models asal mein aag, dhuwan aur insaan pehchaante hain.

Test photo kahan se: aag wali koi bhi photo (JPG/PNG, 5 MB tak). D-Fire dataset ki test photos `Desktop\sentinel-colab\D-Fire.zip` ke andar `test\images\` mein hain (misaal `AoF07831.jpg`). Insaan wali koi bhi photo.

Qadam:

1. Window 1, 2 aur 3 (AI service) chal rahi hon.
2. Upar menu mein **AI** par click.
3. Baen taraf dekhein: hara **ONLINE · all models loaded**, dono models (`person` , `fire_smoke`) ke aage hara nukta.
4. Daen taraf **Choose image...** dabayein aur photo chunein.
5. 1–3 second intezar.



✅ Kya dikhna chahiye:

- [ ] Photo par dabbe: aag **laal**, dhuwan **gray**, insaan **hara**.
- [ ] Neeche line: model ke naam, kitne ms lage, "fire Medium", "smoke yes", "people 0".
- [ ] Table: Class, Confidence (jaise 83%), Box, aur **Why (evidence)** yaani AI ne kyun kaha.
- [ ] Bina aag wali photo par aag ka dabba **na** aaye.

Yaad rakhein:

- Ye sirf **preview** hai: koi incident, alert ya sprinkler **nahi** chalta.
- AI se kabhi galti ho sakti hai. Kuch na mile to iska matlab ye nahi ke jagah mehfooz hai.

Masla:

Screen par	Matlab	Hal
Laal **OFFLINE**	AI service band hai	Window 3 chalayein
Peela "some models missing"	Koi model file nahi	`ai-service\models\` check (Chapter 3, Qadam 4)
"sha256 mismatch"	Model file badli ya kharab	Sahi file dobara copy karein

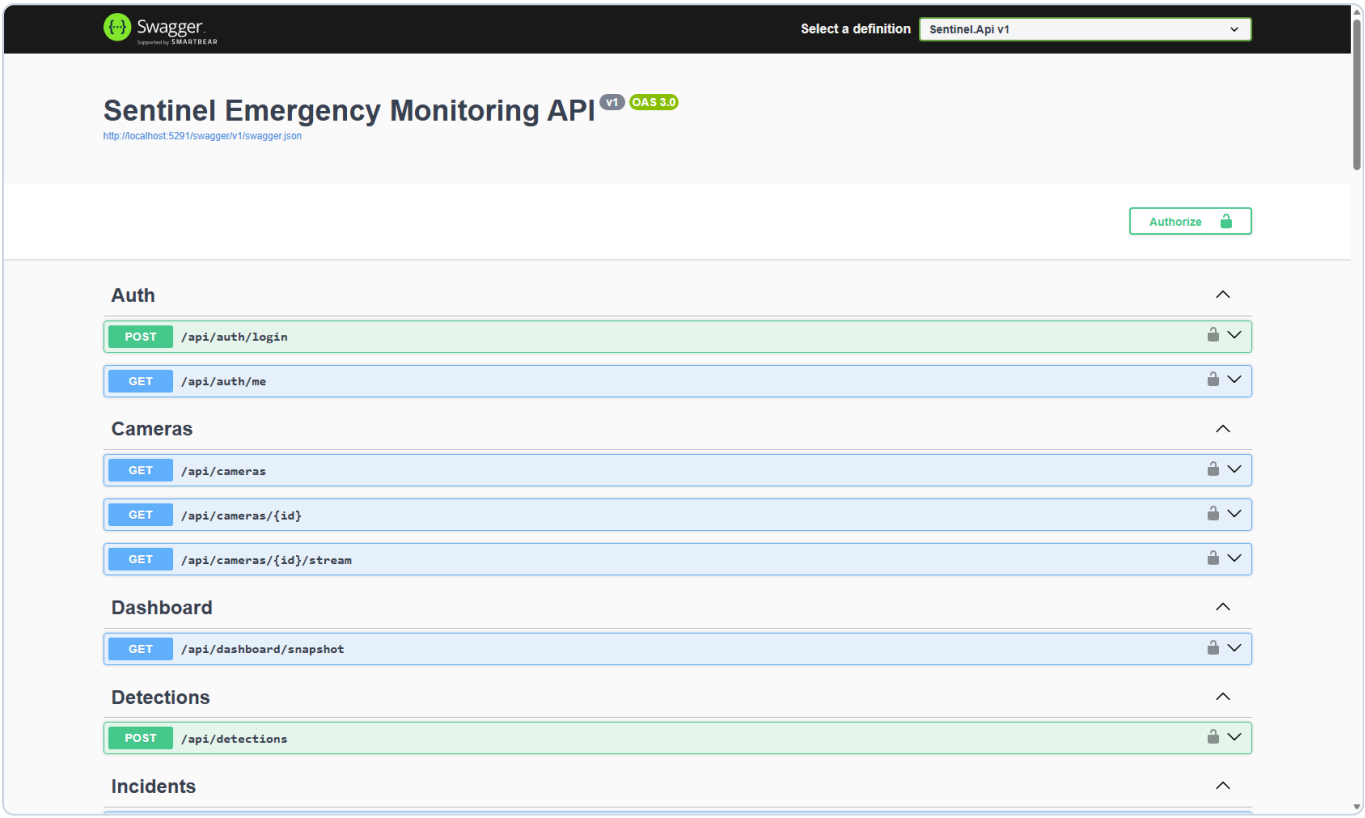
10. Test 3: Sensor Ko Garam Karna (Swagger)

Maqsad: dekhna ke garmi barhne par Warning, Alarm aur Critical sahi waqt par aate hain.

Swagger kya hai? API ka ek test page jahan se bina code ke request bhej sakte hain.

Qadam A: Swagger kholna aur login

1. Browser: **http://localhost:5291/swagger**



2. **Auth** → **POST /api/auth/login** kholein → **Try it out**.
3. Body mein:

```
{ "email": "admin@sentinel.local", "password": "aapka-password" }
```

4. **Execute**. Jawab mein `"token": "eyJ ... "` aayega. Lambi token (quotes ke baghair) copy karein.
5. Upar dayen **Authorize** dabayein → **Bearer** wale dabbe mein token paste → **Authorize** → **Close**.

Qadam B: garmi bhejna

1. **IoT** → **POST /api/iot/sensors/{id}/readings** → **Try it out**.
2. `id` = `temp-b1` (Zone B ka temperature sensor; threshold 57°C).
3. Body mein ek ek karke ye values bhejein, har dafa **Execute**, aur Dashboard dekhein:

Value	Kya hona chahiye	Hisaab
`{ "value": 30 }`	✅ Normal, kuch nahi	45.6 se kam
`{ "value": 50 }`	✅ Gauge **peela** (Warning), Live Event Log mein "Normal → Warning", incident nahi	≥ 57 × 0.8 = 45.6
`{ "value": 60 }`	✅ **High** incident + Zone B ke **Auto** sprinklers ON	≥ 57
`{ "value": 75 }`	✅ **Critical** incident + **emergency** notifications + sprinklers	≥ 57 × 1.25 = 71.25

Sensor IDs:

Zone	Dhuwan (threshold 300 ppm)	Garmi (threshold 57°C)
Zone A	`smoke-a1`	`temp-a1`
Zone B	`smoke-b1`	`temp-b1`
Zone C	`smoke-c1`	`temp-c1`

Dhuwan ke liye Critical = 300 × 1.5 = **450 ppm**.

> ⚠️ Sensor ka data **asli** maana jata hai (SIMULATED nahi), is liye Alarm/Critical par Auto sprinklers **chal jate hain**. Is project mein sprinklers sirf database mein hain, koi asli hardware nahi. Test ke baad unhein OFF karein (Chapter 11).

11. Test 4: Sprinklers

Dashboard → **Smart Sprinklers** panel.

Kaam	Kaise	✅ Kya dikhe
Ek sprinkler ON/OFF	Us ka switch dabayein	Switch badle, Live Event Log mein line
Auto mode	"Auto" button	Alarm/emergency par system khud chala sakta hai
Manual mode	"Manual" button	System khud nahi chalayega, sirf aap
Master	Upar "Master" switch → confirm	Saare sprinklers ek saath ON/OFF

Check: ek sprinkler **Manual** karein, phir Test 3 mein `75` bhejein. ✅ Manual wala sprinkler **OFF** rehna chahiye. Auto wale ON.

Yaad rakhein: sprinkler khud OFF **nahi** hota. Aag ke baad aap OFF karein.

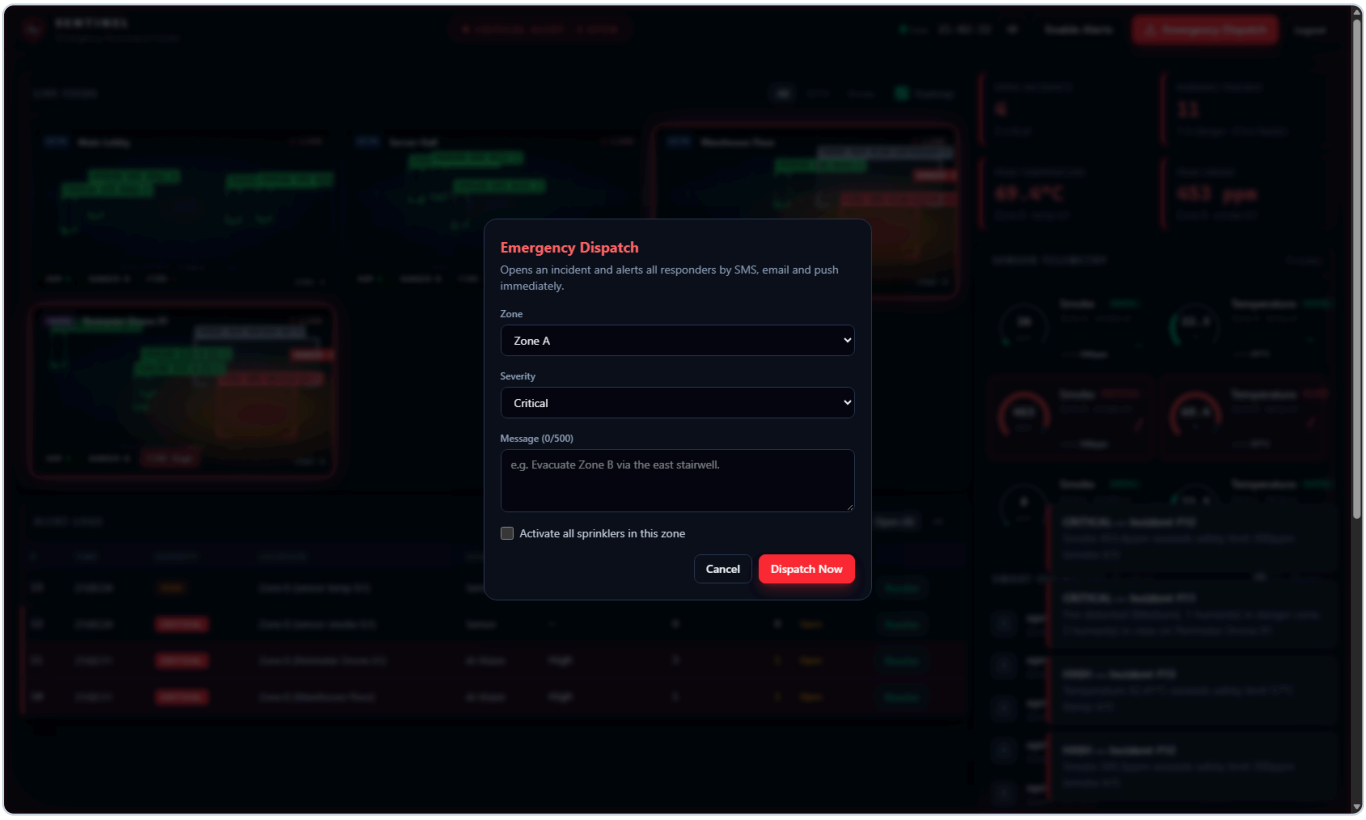
12. Test 5: Emergency Dispatch Aur Resolve

Maqsad: haath se emergency elaan karna, phir incident band karna.

> ⚠️ Dispatch **foran** SMS/Email/Push bhejta hai. Pehle Safe Test Mode (Chapter 4).

Qadam:

1. Upar laal **Emergency Dispatch**.
2. Form:
   - **Zone:** jaise `Zone B`
   - **Severity:** Low / Medium / High / Critical
   - **Message:** jaise "Test: Zone B khaali karein" (500 huroof tak)
   - **Activate all sprinklers in this zone:** tick sirf agar sprinklers chalane hon
3. **Dispatch Now** dabayein.



✅ **Kya dikhna chahiye:**

- [ ] Alert Logs mein naya incident (Source: Manual).
- [ ] Live Event Log: SMS skipped, email API window mein.

Resolve (band karna):

1. **Alert Logs** mein us incident ki line par **Resolve**.

2. ✅ Incident "Open" list se hat jaye ("All" mein Resolved dikhe). History mein `Resolved` log.

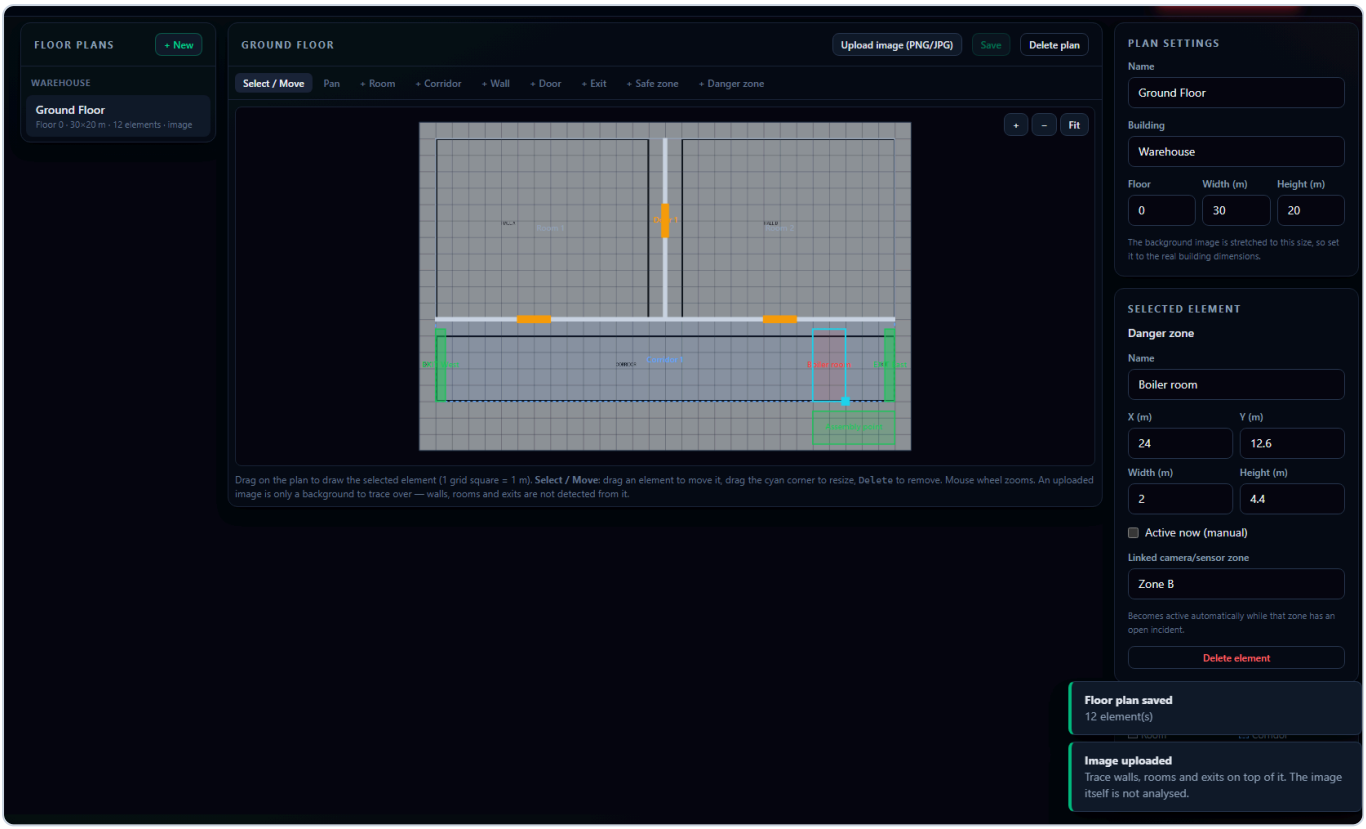
13. Test 6: Naqsha Banana Aur Evacuation Raasta

A. Naqsha banana

1. Upar **Floor Plans** → **+ New**.
2. Daen taraf **Plan settings**: Name ("Ground Floor"), Building ("Warehouse"), Floor (0), **Width aur Height meters mein** (jaise 30 × 20).
3. (Ikhtiyari) **Upload image (PNG/JPG)**: apne naqshe ki photo background mein (5 MB tak).
4. Toolbar se drag karke banayein:

Tool	Kya banana hai
+ Room	Kamre / halls
+ Corridor	Gallery
+ Wall	Deewarein (kamron ke beech)
+ Door	Deewar par darwaza (darwaze ke baghair raasta deewar paar nahi karta)
+ Exit	Bahar nikalne ke darwaze (kam az kam ek)
+ Safe zone	Assembly point
+ Danger zone	Boiler room / khatarnak jagah

5. **Select / Move** se kisi cheez par click → daen panel mein naam, size badlein. Delete key = mitana.
6. Danger zone chunein → **Linked camera/sensor zone** = `Zone B` (Zone B mein incident khula ho to ye khud active).
7. **Save**. ✅ "Floor plan saved" message, title se "unsaved" hat jaye.



Zoom: mouse wheel ya + / − / Fit. Ek khana = 1 meter.

B. Raasta dekhna

1. **Dashboard** → neeche **Evacuation Route (estimate)** panel → floor plan chunein.
2. Naqshe par wahan **click** karein jahan insaan khara hai.
3. ✅ **Hari teer** qareeb tareen exit tak, "Distance 14.8 m · ~12 s" jaisa likha.

C. Aag mein raasta badalna

1. Zone B ke danger zone ke paas wala raasta dekh lein.
2. Test 1 (simulator `--ignite "Zone B"`) ya Test 3 (`temp-b1` = 75) chalayein.
3. ✅ Danger zone laal/active, aur raasta **khud badal** kar doosre exit ki taraf, danger se kam az kam 1 m door.
4. Koi raasta na bache to ✅ saaf message: "No safe route…"

Pehle	Aag ke baad
	

Yaad rakhein: waqt sirf andaza hai (faasla ÷ 1.2 m/s). Dhuwan, bheer, seedhiyan shamil nahi.

D. Camera ko naqshe se jorna (camera ka insaan khud naqshe par)

Maqsad: camera mein dikha insaan naqshe par khud **amber nuqta** ban kar aaye, aur us par click se raasta nikle.

Ek dafa ki tayyari: camera ki ek **snapshot** (photo) jis mein farsh ke 4 kone saaf dikhein, misaal kamre ke 4 kone.

1. **Floor Plans** → naqsha kholein → pehle **Save** (title par "unsaved" na ho).
2. Daen taraf **Camera calibration** → camera chunein (misaal `Warehouse Floor (cam-warehouse)`).
3. Snapshot ko **Paint** mein kholein. Pehle kone par mouse rakhein: Paint neeche x, y pixel dikhata hai. Ye **Camera px** mein likhein.
4. Usi kone ke liye **Plan m** ⌖ dabayein aur naqshe par wohi kona click karein (ya meters khud likhein).
5. Baqi 3 kono ke liye yehi. Koi 3 kone **ek seedhi line** mein na hon.
6. **Save calibration**. ✅ "Camera calibrated" message, aur list mein camera ke aage "calibrated here".



Check:

1. Simulator chalayein (Test 1) ya AI camera (Test 8).
2. Dashboard → **Evacuation Route** → wohi naqsha.
3. ✅ Naqshe par **amber nuqte**: camera ke insaan, unke pairon ki jagah par.
4. Kisi nuqte par click → ✅ **hari teer** usi jagah se exit tak.



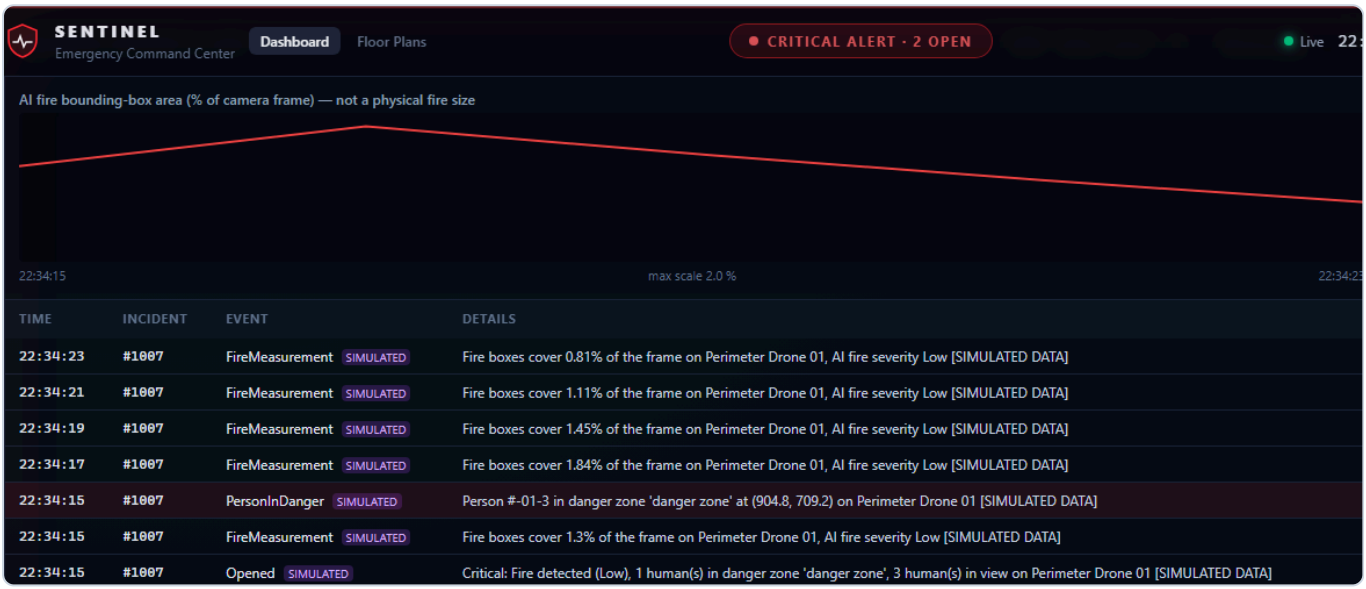
Masle:

Screen par	Matlab	Hal
"Three of the 4 points lie on one line"	3 kone ek line mein	Alag alag kone chunein
"A point lies outside…"	Naqshe se bahar ka point	Naqshe ke andar ke meters
"Save the plan first"	Naqsha save nahi	Pehle **Save**
Nuqte nahi aate	Camera se frames nahi, ya ghalat naqsha chuna	Simulator/AI chalayein, wohi naqsha chunein
Nuqte ghalat jagah	Points ki tarteeb ulti (camera ka kona 1 ≠ naqshe ka kona 1)	Dono taraf **ek hi kona** ek hi number par

> ⚠️ Camera **hila** to dobara calibrate karein. Nuqta andaza hai, aur seedhiyon par khade insaan ke liye ghalat ho sakta hai.

14. Test 7: Incident History Aur Logs

1. Dashboard → **Incident History & Logs**.
2. Upar se ek **incident** chunein (ya "All incidents"), aur event type ("All event types").
3. ✅ Dikhe:
   - [ ] Camera wale incident par **aag ka graph** (frame ka %), sensor wale par **sensor ka graph**.
   - [ ] Logs: `Opened`, `Escalated`, `FireMeasurement`, `PersonInDanger`, `EvacuationRoute`, `Resolved`.
   - [ ] Test data par jamni **SIMULATED**, raaste par neela **ESTIMATE**.



15. Test 8: AI Ko Video Se Camera Ki Tarah Chalana

Maqsad: AI ko lagatar frames dena, jaise asli camera, aur dekhna ke wo khud incident banata hai.

Qadam:

1. Aag ki ek **video file** (MP4/AVI) computer par rakhein.
2. `ai-service\config.example.json` ki copy banayein, naam `config.json`.
3. Notepad mein `cameras` hissa aisa karein (apna path likhein, `\` ki jagah `/`):

```
{
  "api": "http://localhost:5291",
  "listen": "127.0.0.1:8008",
  "cameras": [
    { "cameraId": "cam-warehouse", "source": "C:/Users/m_fahad/Videos/fire-test.mp4", "fps": 2,
"loop": false, "enabled": true }
  ]
}
```

4. Window 3 band karein (Ctrl + C), phir device key de kar dobara chalayein:

```
cd C:\Users\m_fahad\Desktop\Project\ai-service
$env:SENTINEL_DEVICE_KEY = "<yahan Device__ApiKey>"
.\.venv\Scripts\python -m sentinel_ai
```

✅ **Kya dikhna chahiye:**

- [ ] AI page → Camera sources: `cam-warehouse · file · SIMULATED / TEST INPUT · running · sent …`
- [ ] Dashboard ke Warehouse camera par AI ke boxes.
- [ ] Aag **3 frames** tak lagatar dikhe to incident, **SIMULATED DATA** label ke saath.
- [ ] Sprinklers **khud nahi** chalte (simulated).

Yaad rakhein:

- Video **file** hamesha SIMULATED maani jati hai.
- Agar video mein manzar har frame badalta ho (photos ka slideshow), to aag "confirm" nahi hogi. `fps` barha kar (jaise 4) dobara try karein.
- Is purane computer par kul ~1–2 frame per second ka budget hai.
- **Asli camera** (`rtsp://…`) jorne se pehle Safe Test Mode aur ijazat zaroori. Camera ka password config mein nahi, environment variable mein rakhein.

16. Test 9: Browser Notifications

1. Upar **Enable Alerts** dabayein → browser poochhe to **Allow**.
2. Test 1 ya Test 5 chalayein.
3. ✅ Windows ke kone mein Sentinel ka notification aaye (dashboard ka tab band ho tab bhi, jab tak browser background mein chal raha ho).

Ye sirf tab kaam karta hai jab `.env` mein `Notifications__WebPush__…` keys bhari hon. Keys banane ki command:

```
cd C:\Users\m_fahad\Desktop\Project\backend
dotnet run --project Sentinel.Api -- generate-vapid-keys
```

17. Automatic Tests (Ek Click Mein Sab Check)

Ye tests code khud check karte hain. Har command ke aakhir mein **failed: 0** hona chahiye.

```
cd C:\Users\m_fahad\Desktop\Project\backend
dotnet test                                  # ✅ Passed: 56

cd C:\Users\m_fahad\Desktop\Project\frontend
ng test --watch=false                        # ✅ Tests 25 passed
ng build                                     # ✅ "Application bundle generation complete"

cd C:\Users\m_fahad\Desktop\Project\ai-service
.\.venv\Scripts\python -m unittest discover -s tests    # ✅ Ran 15 tests ... OK
```

AI ki accuracy dobara napna (D-Fire test folder ka path chahiye, ~20 minute):

```
.\.venv\Scripts\python tools\evaluate.py --data <D-Fire\test folder> --classes 0=Smoke,1=Fire --
sweep 0.25,0.3,0.35,0.4,0.5 --out reports\meri-jaanch
```

✅ Threshold 0.4 par fire image recall ~0.93 aur smoke ~0.92 aana chahiye.

18. Masle Aur Un Ka Hal

Masla	Wajah	Hal
Top bar par **Offline / Reconnecting**	API band hai	Window 1 chalayein
Login par "network error"	API band, ya port 5291 mein masla	Window 1 check, `http://localhost:5291/health` kholein: `"status":"ok"` aana chahiye
Ghalat password	`.env` ka password	`.env` mein `Seed__AdminPassword` dekhein. Users table pehle se ho to wohi purana password
API: "Cannot connect to SQL Server"	SQL Server band	Start → "Services" → **SQL Server (MSSQLSERVER)** → Start
API: "pending migration(s)"	Database purana	`dotnet ef database update --project Sentinel.Api` (Chapter 3)
API error: "Invalid column name 'Calibration'"	Naya update (camera calibration) lag gaya, database purana	Wohi command: `dotnet ef database update --project Sentinel.Api`
Cameras par **NO AI DATA**	AI/simulator data nahi bhej raha	Simulator (Test 1) ya AI camera (Test 8) chalayein
AI page **OFFLINE**	AI service band	Window 3 chalayein
AI "some models missing"	`.onnx` file nahi	Chapter 3, Qadam 4
AI camera "API HTTP 401"	Device key ghalat ya nahi di	`$env:SENTINEL_DEVICE_KEY` = `.env` ka `Device__ApiKey`
Simulator "401"	Device key ghalat	`--device-key` mein sahi key
`ng` / `dotnet` "not recognized"	Software install nahi ya PowerShell purani	Chapter 2, phir **nayi** PowerShell kholein
Port "already in use"	Pehle se chal raha hai	Purani window band karein, ya computer restart
SMS/Email nahi aate	Safe Test Mode on, ya settings ghalat	`#` hatayein (Chapter 4), Live Event Log mein error dekhein
Floor plan save par laal message	Koi cheez naqshe se bahar	Message mein naam likha hota hai, us ko andar karein
Raasta: "Start point is not inside..."	Click kamre/corridor se bahar	Room/Corridor ke andar click karein

19. Kya Kabhi Na Karein

❌ Na karein	Kyun
Ise asli fire alarm ki jagah lagana	Ye test aur lab mein check hua hai, certified nahi
Bina Safe Test Mode ke aag ka test	Asli logon ko asli SMS/Email chale jayenge
Asli sprinkler/alarm hardware bina ijazat jorna	Paani ya ghabrahat se nuqsan
`.env` kisi ko bhejna ya internet par daalna	Is mein passwords aur keys hain
AI ke "kuch nahi mila" ko "sab mehfooz" samajhna	AI se galti ho sakti hai
Camera ka password `config.json` mein likhna	Environment variable mein rakhein
Model files (`.onnx`) badalna	Fingerprint match nahi hoga, model band ho jayega

Yaad rakhein: aag mein bijli jaye to camera band ho sakte hain. Cameras ko UPS/PoE backup par rakhein, aur hamesha smoke detectors aur fire panel istemaal karein.

20. Test Checklist (Print Karke Tick Karein)

#	Test	Chapter	✅ / ❌	Note
1	Software install, versions dikhe	2		
2	`.env`, database, packages, AI setup	3		
3	Safe Test Mode on	4		
4	Teeno windows chalin (5291, 4200, 8008)	5		
5	Login hua, "Live" hara	6		
6	Simulator: Zone B mein aag, CRITICAL, incident	8		
7	AI page: aag ki photo par laal dabba	9		
8	AI page: bina aag photo par aag nahi	9		
9	Sensor 50 = Warning, 60 = High, 75 = Critical	10		
10	Sprinkler: Manual wala khud nahi chala	11		
11	Emergency Dispatch + Resolve	12		
12	Naqsha bana aur save hua	13		
13	Raasta + waqt dikha	13		
14	Aag mein raasta khud badla	13		
14b	Camera calibrate hua, naqshe par amber nuqte, nuqte se raasta	13		
15	History mein graph aur logs	14		
16	AI video camera: SIMULATED incident	15		
17	Browser notification aaya	16		
18	Automatic tests: 56 + 25 + 15 pass	17		
19	Safe Test Mode **band** kiya (`#` hataye)	4		

Tester ka naam: ..... **Tareekh:** .....